

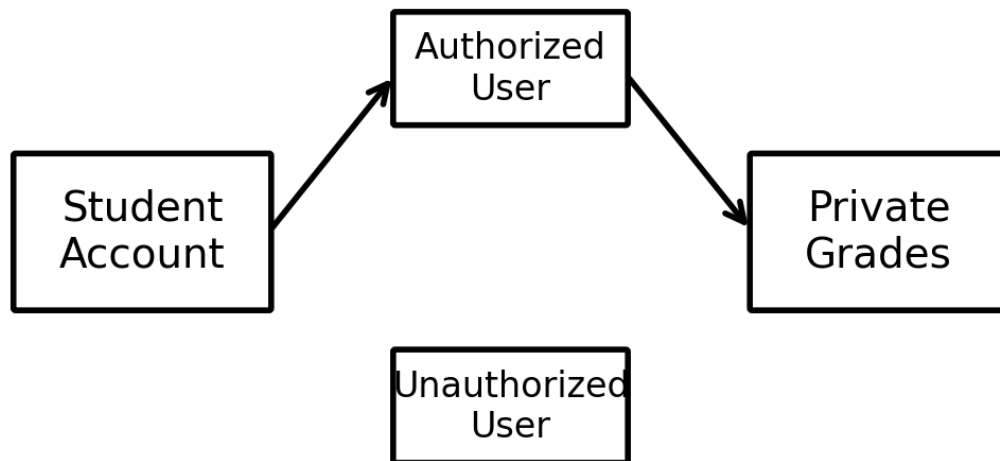
Task 2: Core Cybersecurity Concepts Explainer

Beginner-friendly explanations with original illustrations

This document explains five core cybersecurity concepts in plain language. Each concept has one original illustration created specifically for this document, along with a concrete real-world scenario and an explanation of what happens when the concept is violated.

1. Confidentiality

Original Illustration 1 — Confidentiality



Confidentiality: only approved people can view private information.

What it means: Confidentiality means keeping information private so that only people who are allowed to see it can access it.

Real-world scenario: Imagine a college student portal containing a student's marks. The student's grades should be visible to the student and approved college staff, but not to random students. Login controls and permissions help prevent unauthorized people from reading the grades.

What happens if it is violated: If confidentiality is violated, private information can be exposed. For example, if one student can open another student's grade page, the private academic information is no longer confidential.

Key takeaway: Confidentiality is about preventing unauthorized viewing or disclosure.

2. Integrity

Original Illustration 2 — Integrity



Unauthorized change: 90/100 → 100/100 ×

Integrity means data stays accurate unless an authorized change is made.

What it means: Integrity means keeping information correct, complete, and protected from unauthorized changes.

Real-world scenario: Suppose a lecturer records a student's assignment mark as 90/100. The stored mark should remain 90/100 unless an authorized lecturer changes it for a valid reason.

What happens if it is violated: If someone changes the mark to 100/100 without permission, the data is no longer trustworthy. Permissions, audit logs, and integrity checks can help prevent or detect unauthorized changes.

Key takeaway: Integrity is about keeping data accurate unless an authorized change is made.

3. Availability

Original Illustration 3 — Availability



Server failure → portal unavailable → submission blocked

Availability means authorized users can access a service when needed.

What it means: Availability means that systems and information should be accessible to authorized users when they need them.

Real-world scenario: A college learning portal should be available when students need to download notes or submit an assignment. If the server is unavailable during the submission period, students may not be able to complete their work.

What happens if it is violated: Availability can be affected by hardware failures, software problems, network failures, or denial-of-service attacks. Backups, redundancy, maintenance, and recovery plans can help keep services available.

Key takeaway: Availability is about making sure the service is there when it is needed.

4. Authentication vs Authorization

Original Illustration 4 — Authentication vs Authorization



Example: a student may view their own grades but cannot edit everyone's grades.

What it means: Authentication answers 'Who are you?' Authorization answers 'What are you allowed to do?' They are related, but they are not the same.

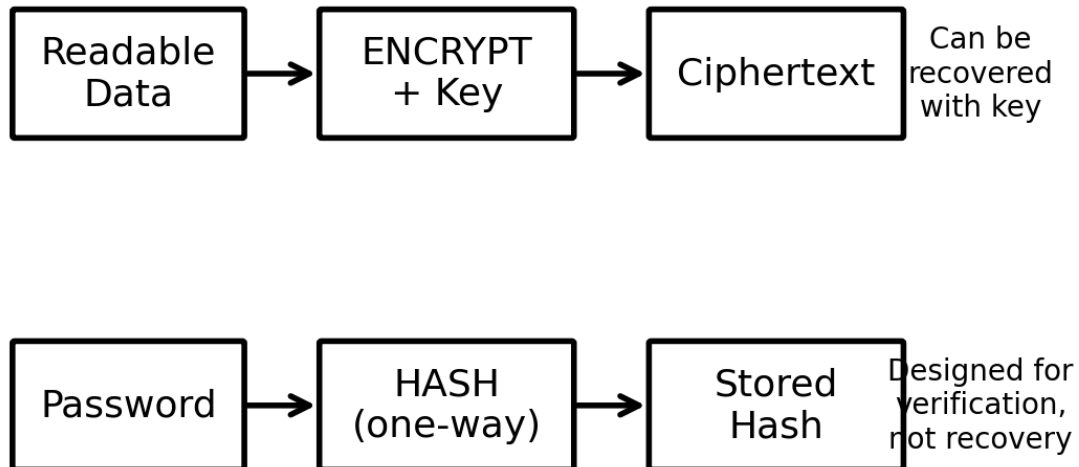
Real-world scenario: When a student signs into a college portal, the login process authenticates the student. After login, authorization may allow the student to view their own grades but not edit the grades of other students.

What happens if it is violated: If a system authenticates a person but gives them excessive permissions, they may access information or actions they should not have. Least privilege means giving users only the access they need.

Key takeaway: Authentication checks identity; authorization checks permissions.

5. Encryption vs Hashing

Original Illustration 5 — Encryption vs Hashing



What it means: Encryption changes readable data into ciphertext so it can later be recovered with the appropriate key. Hashing converts data into a fixed-length value and is designed to be one-way.

Real-world scenario: A banking application may encrypt sensitive information while it is transmitted or stored so an authorized system can decrypt it later. Passwords, however, should normally be stored using a secure password-hashing method rather than reversible encryption.

What happens if it is violated: If encrypted data is intercepted without the key, the attacker should not be able to read the original message easily. With password hashing, the goal is different: the original password should not be recoverable from the stored hash.

Key takeaway: Encryption protects data that may need to be recovered; password hashing is used to verify passwords without storing the original password.

How the Five Concepts Work Together

A secure college portal can use all five ideas together: authentication verifies identity; authorization limits what a user can do; confidentiality protects private information; integrity protects the accuracy of stored information; and availability keeps the portal usable when authorized users need it. Encryption and secure password hashing are technical controls that

support these security goals.

Original Illustrations Note

All five diagrams in this document were newly drawn for this submission. They are original illustrations created to explain the five concepts and are not recycled stock images.